

Data Structures: Unit 1 Dynamic Memory and Linked Lists

VTU 2022 Scheme CBCS - BCS303 Module 1

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Dynamic Memory Allocation in C: malloc, calloc, realloc, and free functions
2. Singly Linked List: Node definition, Insertion, Deletion, and Traversal operations
3. Circular Linked List: Header nodes and round-robin scheduling applications
4. Doubly Linked List: Left/Right pointers and two-way traversals
5. Polynomial Addition and Linked Stack/Queue implementations

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Write complete C function to insert node at specific position in SLL [10 Marks]
- Q2. Demonstrate polynomial addition using singly linked list [10 Marks]